Lab 15: Macroevolution (Module 14)

BIOL-1

Name: ______________________ **Date:** ______________________

Learning Objectives

By the end of this lab, you should be able to:

1. Apply the biological species concept to **predict** when **gene flow** can stop or reverse lineage splitting.
2. Classify **prezygotic** vs **postzygotic** barriers and name sub-types from **messy** narratives.
3. Contrast **allopatric** vs **sympatric** speciation with **plausible** mechanisms.
4. Explain how microevolutionary change can add up to **reproductive isolation**.

Overview

Where this sits: In **Module 13 / Lab 14**, you followed **allele frequencies** in **one** population (microevolution). In **Module 14 / Lab 15**, the question shifts: **When does one lineage become two?** Reproductive **isolation** and **reduced gene flow** are the stars.

You are on the **Speciation Response Team**. A river canyon has split one ancestral **mock-orange fly** (fictional species for this worksheet) into **Population North** and **Population South**. Your job is to **predict** what must happen for **one species to become two**, then diagnose **barriers** from field reports, and finally decide whether a weird plant counts as **quick speciation**. This lab is a chain of **if–then** puzzles on paper—not a vocabulary list to memorize out of order.

Terms you will use (one glance)

Term	In one line
Biological species concept (BSC)	Groups are often treated as one species if they **interbreed** (or could) and produce viable/fertile offspring; **ongoing gene flow** tends to keep them one line.
Reproductive isolation	**Barriers** that make **interbreeding** rare or **hybrids** unsuccessful—**prezygotic** (before a zygote) or **postzygotic** (after).
Prezygotic	

Term	In one line
	Mating, timing, or gametes “don’t meet” the way you’d need for a zygote (first line of **no gene flow**).
Postzygotic	A zygote or hybrid has trouble surviving or **reproducing** (e.g. embryo dies, sterile hybrid).
Allopatric	Isolation that starts with a **geographic** split.
Sympatric	Speciation in the **same** area (e.g. **instant** isolation from a genome change in one plant).

Materials Needed

- Pencil or pen
- This worksheet

Part 1: Canyon Split — Build a Speciation Hypothesis

One species, two valleys, no bridges. Flies rarely cross the canyon.

Road map: A. Predict (before the “report”) → **B. Data** (read the field report) → **C. Revise** (fix your first idea with evidence).

A. Prediction: If North and South stay isolated for thousands of generations, **mutations and selection** can proceed independently. **Before you read “Report B,”** answer:

Hypothesis: Will the two populations *necessarily* become two separate species **automatically** just because they end up **looking** different? **Yes / No.**
Hint: The biological species concept is about mating, offspring, and gene flow—not about looks alone. Looking different = morphology; it does not, by itself, “clock in” as two BSC species.

Reason:

B. Report (given): After 10,000 generations, North flies breed in **early spring**; South flies breed in **late spring**. When researchers **cage** them together, mating is **rare**; when they **force** mating, eggs are fertilized but **embryos die** before hatching.

1. List two distinct reproductive barriers in that report (name the **types**—e.g. temporal, reduced hybrid viability, etc.):

1.

2.

2. Which barrier is prezygotic? Which is postzygotic?

- Prezygotic:
- Postzygotic:

C. Revision: Did your Part A idea need to change? What **evidence** made you tighten or fix it?

Part 2: Mystery Reports — Diagnose the Barrier

Each story is incomplete on purpose—your call on Pre vs Post and sub-type is the point.

3. For each report, mark **Pre** (prezygotic) or **Post** (postzygotic).

Part 2 — Prezygotic or postzygotic

#	Field report	Pre or Post?
1		
2		
3		
4		

4. Sub-type: Match each line of **question 3** to the best label. Choose from: *habitat* / *temporal* / *behavioral* / *mechanical* / *gametic* / *reduced hybrid viability* / *reduced hybrid fertility* / *hybrid breakdown*.

Part 2 — Barrier sub-type

#	Report (same order as in Q3)	Barrier sub-type
1		
2		
3		
4		

Stretch (Part 2). Short answers—use the same four reports if helpful.

(a) In one sentence: why does **behavioral isolation** (e.g. different firefly flash patterns) block gene flow **before** a zygote counts as "formed" for the pre/post split?

(b) The **F1 salamanders** that survive but do not mate successfully: state **prezygotic or postzygotic** and the **sub-type** label you would use.

(c) Suppose hybrids between two plant species **germinate** but **die** as seedlings before they flower. Is the main barrier **pre-** or **postzygotic**, and why?

Part 3: Geography — Same River, Different Endings

Polyploidy in one line: A cell can get **extra full sets of chromosomes** (e.g. **4n** has twice the "library" of a typical **2n**). A new **4n** plant in the same meadow is often **instantly reproductively isolated** from **2n** relatives, so a new **species** can form **without** a geographic split.

5. Two roads to new species:

Part 3 — Allopatric or sympatric

#	Scenario	Allopatric or Sympatric?
1		
2		

6. Mechanism check: Why is polyploidy often **sympatric** speciation in plants? *(Hint: "instant" isolation from the diploid population in the same field.)*

7. Counterfactual: If the canyon **fills in** and the flies **mix** freely **before** strong barriers evolve, what happens to **this** speciation story? **One paragraph (4–5 sentences):**

Part 4: From Lab 14 to Today — Micro Builds Macro

No new "evolution force" appears at the speciation line—barriers to gene flow are what you add to the microevolution you already know.

8. Chain argument: Starting from **mutation** creating variation in each isolated population, explain how **natural selection** and **drift** could make **prezygotic** barriers more likely over time. (4–5 sentences)

9. Final verdict: Under the biological species concept, if two groups still have **fertile hybrids** whenever they meet, are they **definitely** two separate species? **Gene flow** vs **diverging lineages** in 3–4 sentences.

10. Pace of splitting: Which path in this lab—**many generations** in separated valleys (**allopatric** mock-orange flies) or **one-generation** polyploidy in the **same** meadow—typically needs **more time** for **independent** evolution before the groups are reproductively isolated? **One short paragraph.**

11. BSC limits: Name **one situation** where the **biological species concept** is **hard to apply** in the field (hint: fossils-only lineages, bacteria, or strictly asexual groups). **Two or three sentences.**

Wrap-up

W1. Name **one** barrier you almost mixed up (pre vs post, or two sub-types) and what finally **clicked**.

W2. Optional fun: One **real** or **movie/TV** example of **isolation** (courtship, timing, or hybrid trouble)—**one** sentence, no need for perfect Latin names.

W3. Looks aren't enough: When would using **appearance alone** be a **misleading** way to decide if two populations are **one** species or **two**?

W4. Micro → macro: In **Lab 14** you tracked **allele frequencies** in one population. If **migration** between North and South mock-orange flies were **zero** for a long time, how does that help the **two valleys** accumulate **different** genetic changes? **2–3 sentences.**
